Bayesian Framework for Canonical factorization of multiblock data

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1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data commonly arise when several sets of variables are collected on the same individuals to capture different psychological domains, such as self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

Exploratory multiblock data analysis has four main objectives: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. A unified algebraic framework based on factorization strategies, called *canonical factorization*, was proposed by Bougeard et al. (2024) for multiblock methods to achieve these objectives. In practice, canonical factorization provides a structured framework for analyzing data variability at multiple levels: differences between and within blocks, and differences between observations through their projection onto a common subspace. It also facilitates the harmonization of the outputs of different methods and supports a criterion-based selection among competing approaches.

The canonical factorization is exact when the dimension of the common subspace equals the rank of the concatenated matrix, which can be large. Determining a reduced dimension, and identifying which dimensions are common to all blocks, shared by some of them or specific to one block, is therefore a major issue. Bayesian approaches address the first question for a single matrix: Hoff (2007) proposed a Bayesian singular value decomposition (SVD) that quantifies the uncertainty on the rank, and competing approaches were obtained with other priors on the singular values, such as the double exponential distribution Zhai and Gutman (2023), the repulsed normal distribution Yuchi et al. (2023) and the horseshoe prior Wang et al. (2026), which shrinks small singular values towards zero.

Following this line of work, we propose a Bayesian framework for the canonical factorization of multiblock data. The model imposes orthonormality constraints on both the common components and the block loadings, which makes the axes of the common subspace identifiable and leads to matrix von Mises–Fisher full conditional distributions. We derive an efficient Gibbs sampler and show that its conditional modes coincide with the updates of Optimal Multiple Co-Inertia Analysis (OMCOA), the simultaneous alternative to Multiple Co-Inertia Analysis Chessel and Hanafi (1996). We then propose a sparse version based on a spike-and-slab prior Mitchell and Beauchamp (1988); George and McCulloch (1993) on the block weights. Unlike continuous shrinkage priors, this prior yields exact zeros and posterior inclusion probabilities, which directly provides an estimate of the dimension of the common subspace.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals,
- $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks,
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$,
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm,
- $\mathcal{V}_{N,R} = \{\mathbf{W} \in \mathcal{M}_{N,R}(\mathbb{R}) : \mathbf{W}^\intercal \mathbf{W} = \mathbf{I}_R\}$ denotes the Stiefel manifold.

Consider K column blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively in $\mathcal{M}_{N,p_1}(\mathbb{R}), ..., \mathcal{M}_{N,p_K}(\mathbb{R})$. Let $\mathbf{X} = [\mathbf{X}_1 | \ldots | \mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ $(P = \sum_{k=1}^{K} p_k)$ be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

Bougeard et al. (2024) demonstrated that the $(K+1)-$blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = rank(\mathbf{X})$ as follows

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \ldots, K) \\ \text{with the constraints} \quad \mathbf{U} \in \mathcal{V}_{N,R} \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \ldots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \ldots, K), \end{cases} \tag{1}$$

where

- $\mathbf{U} \in \mathcal{V}_{N,R}$ is the block-common components,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ are the block-loadings, and $\mathbf{V} = [\mathbf{V}_1 | ... | \mathbf{V}_K] \in \mathcal{M}_{P,R}(\mathbb{R})$,
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^{K} \Sigma_k^2$ that relate common and block-analyses.

Without loss of generality, we can assume $\sigma_k^{(r)} \geq 0$ for all k, r, since a joint sign change of $\sigma_k^{(r)}$ and $\mathbf{v}_k^{(r)}$ leaves the model and the constraints unchanged. Since the blocks $\mathbf{X}_1, ...\mathbf{X}_K$ are column-centered, the block-common $\mathbf{U} \in \mathcal{V}_{N,R}$ is centered as-well:

$$\mathbf{U} \in \mathcal{V}_{N,R}, \quad \mathbf{1}_N\mathbf{U} = 0.$$

This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see Bougeard et al. (2024)). For all the multiblock methods, once $\mathbf{U}$ is fixed, the decomposition $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$ of each block is unique (up to sign). The methods differ in how the axes $\mathbf{u}^{(1)}, \ldots, \mathbf{u}^{(R)}$ themselves are determined. When the block loadings are only normalized ($\|\mathbf{v}_k^{(r)}\| = 1$), the optimal loadings are $\mathbf{v}_k^{(r)} = \mathbf{X}_k^\intercal\mathbf{u}^{(r)} / \|\mathbf{X}_k^\intercal\mathbf{u}^{(r)}\|$ and the least-squares criterion reduces to

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}\|^2 - \|\mathbf{X}^\intercal\mathbf{U}\|^2,$$

which is invariant under any rotation $\mathbf{U} \mapsto \mathbf{UR}$: only span($\mathbf{U}$) is determined by the data, and the axes have to be fixed by an external convention, usually the SVD of $\mathbf{X}$. This convention becomes unstable when the singular values of $\mathbf{X}$ are close. When $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all k, the criterion is no longer rotation invariant, and the axes of $\mathbf{U}$ are identified (up to permutation and sign) by the block-specific profiles $\Sigma_1, \ldots, \Sigma_K$, as long as these profiles differ across blocks.

2.1 Canonical Factorisation under orthogonal block-loadings constraint

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for $k = 1, \ldots, K$. Chessel and Hanafi Chessel and Hanafi (1996) proposed a sequential solution, known as Multiple Co-Inertia Analysis (MCOA): the dimensions are computed one at a time, each from the blocks deflated with respect to the previous block loadings. This procedure yields orthonormal matrices $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$, and each stage is optimal in the least-squares sense given the previous ones. The resulting solution is nevertheless suboptimal with respect to (2), since each dimension is chosen without taking the next ones into account, so that the R dimensions are not fitted simultaneously. Optimal Multiple COinertia Analysis (OMCOA) is proposed as a simultaneous solution of (2) under the constraints $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ ($k = 1, \ldots, K$). It is computed by an alternating least squares (ALS) algorithm, in which each step minimizes (2) with respect to one set of parameters, the others being fixed:

- **Update of U:** $\mathbf{U} = \mathbf{AB}^\intercal$, where $\mathbf{XH}^\intercal = \mathbf{ADB}^\intercal$ is the SVD of $\mathbf{XH}^\intercal$, with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$ (orthogonal Procrustes problem);
- **Update of $\mathbf{V}_k$:** $\mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$, for $k = 1, \ldots, K$;
- **Update of Σ_k:** $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$, for $k = 1, \ldots, K$.

Proposition 1. *The successive updates of $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ guarantee that the criterion defined in* (2) *decreases or remains constant at each step, ensuring convergence of the algorithm.*

The proof is given in the appendix.

Remark 1. *Since the limit may however be a local minimum, the algorithm is started from the MCOA solution and from several random initializations, and the solution with the lowest criterion is retained.*

Proposition 2 (Noiseless identifiability)**.** *Let $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, $k = 1, \ldots, K$, with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$, $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, $\sigma_k^{(r)} \geq 0$, and assume that every dimension is present in at least one block: for each r there is a k with $\sigma_k^{(r)} > 0$. Denote by $\mathbf{s}^{(r)} = (\sigma_1^{(r)}, \ldots, \sigma_K^{(r)})$ the profile of dimension r across the blocks and, for block weights $w_1, \ldots, w_K > 0$, by*

$$\lambda_r(\mathbf{w}) = \sum_{k=1}^{K} w_k (\sigma_k^{(r)})^2$$

the squared singular values of the weighted concatenated matrix $[\sqrt{w_1}\mathbf{X}_1 | \cdots | \sqrt{w_K}\mathbf{X}_K]$.

(a) ***OMCOA.*** *If the profiles are pairwise distinct, $\mathbf{s}^{(r)} \neq \mathbf{s}^{(r')}$ for $r \neq r'$, then every minimiser of criterion (2) in dimension R, weighted or not, is equal to $(\mathbf{U}, \mathbf{V}_k, \Sigma_k)$ up to a common permutation of the dimensions and to the joint change of sign $(\mathbf{u}^{(r)}, \mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)}) \mapsto -(\mathbf{u}^{(r)}, \mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)})$; the columns $\mathbf{v}_k^{(r)}$ with $\sigma_k^{(r)} = 0$ are not determined.*

(b) ***MCOA.*** *If the values $\lambda_1(\mathbf{w}), \ldots, \lambda_R(\mathbf{w})$ are pairwise distinct, MCOA applied to the weighted blocks $\sqrt{w_k}\mathbf{X}_k$ recovers $(\mathbf{U}, \mathbf{V}_k)$ up to the order and the signs of the dimensions, and the weights $\sqrt{w_k}\,\sigma_k^{(r)}$. Its solution then coincides with the OMCOA solution (once the MCOA weights are divided by $\sqrt{w_k}$).*

(c) *If $\lambda_r(\mathbf{w}) = \lambda_{r'}(\mathbf{w})$ for some $r \neq r'$, the MCOA solution is not determined by the data: at the first step where the largest remaining value is tied, any unit vector of the corresponding eigenspace is a solution, and the computed solution differs in general from $(\mathbf{U}, \mathbf{V}_k)$, even when the profiles are distinct.*

Since equal profiles give equal values $\lambda_r(\mathbf{w})$ for every $\mathbf{w}$, the condition of (b) implies that of (a), whatever the weights. Conversely, for any weights there are designs with distinct profiles and tied values $\lambda_r(\mathbf{w})$: OMCOA identifies strictly more configurations than MCOA, and its identifiability does not depend on the block weighting.

3 Bayesian framework for Canonical Factorisation

A probabilistic counterpart of (2) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \sim \mathcal{N}(0, \tau_k^{-1}), \qquad k = 1, \ldots, K, \tag{3}$$

where τ_k is the noise precision of block k. centered as well: $\mathbf{U} \in \mathcal{V}_{N,R}$ with $\mathbf{1}_N^\intercal\mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative von Mises–Fisher distribution). A random matrix $\mathbf{W} \in \mathcal{V}_{m,R}$ follows the matrix von Mises–Fisher distribution vMF($\mathbf{F}$), with parameter $\mathbf{F} \in \mathcal{M}_{m,R}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,R}$ is proportional to $\exp\{\mathrm{trace}(\mathbf{F}^\intercal\mathbf{W})\}$ Hoff (2009). If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the SVD of $\mathbf{F}$, the mode of vMF($\mathbf{F}$) is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.
- $\sigma_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, a half-normal distribution with a small precision λ (e.g. $\lambda = 10^{-2}$);
- $\tau_k \sim \mathrm{Gamma}(a_0, b_0)$ with small hyperparameters (e.g. $a_0 = b_0 = 10^{-3}$).

3.1 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\left\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\right\|^2 = \|\mathbf{X}_k\|^2 - 2\,\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) + \sum_{r=1}^{R}(\sigma_k^{(r)})^2. \tag{4}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood is

$$\log p(\mathbf{X}_1, \dots, \mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K} \Big[\tfrac{(N-1)p_k}{2} \log \tau_k - \tfrac{\tau_k}{2} \big\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \big\|^2 \Big] + \text{const.}$$

By (4), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\operatorname{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Proposition 3 (Full conditionals)**.** *Under model* (3) *and the above priors:*

1. ***Common space.*** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ with $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,R}$. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{Q}^\intercal \mathbf{F}_\mathbf{U}), \qquad \mathbf{F}_\mathbf{U} = \sum_{k=1}^{K} \tau_k \, \mathbf{X}_k \mathbf{V}_k \Sigma_k.$$

2. ***Block loadings.*** *For $k = 1, \dots, K$,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k}), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k \, \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k.$$

3. ***Block weights.*** *For $k = 1, \dots, K$ and $h = 1, \dots, H$, let $d_k^{(r)} = \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$. Then*

$$\sigma_k^{(r)} \mid \cdot \;\sim\; \mathcal{N}^+\Big(\frac{\tau_k \, d_k^{(r)}}{\tau_k + \lambda}, \; \frac{1}{\tau_k + \lambda} \Big),$$

   a normal distribution truncated to $(0, +\infty)$.

4. ***Noise precisions.*** *For $k = 1, \dots, K$,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k}{2}, \; b_0 + \frac{1}{2} \big\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal \big\|^2 \Big).$$

The modes of the full conditionals of Proposition 3 are

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_\mathbf{U} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k \mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k \mathbf{D}_k \mathbf{B}_k^\intercal, \qquad \sigma_k^{(h)} = \frac{\tau_k}{\tau_k + \lambda} \, d_k^{(h)}.$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \cdots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1 \mathbf{V}_1^\intercal | \cdots | \Sigma_K \mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \operatorname{diag}(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)$. OMCOA is therefore the iterated conditional modes (ICM) algorithm Besag (1986) associated with this posterior, and the Gibbs sampler is its stochastic counterpart. With block-specific precisions, $\mathbf{F}_\mathbf{U} = \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion (2), a common precision $\tau_1 = \cdots = \tau_K = \tau$ can be used, with (2), $\tau \mid \cdot \sim \mathrm{Gamma}\big(a_0 + \sum_k (N-1)p_k/2, \; b_0 + \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2/2\big)$.

Post-processing: label switching and signs The posterior distribution is invariant under any permutation π of the R dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \ldots, \mathbf{V}_K$ and $\Sigma_1, \ldots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(r)}$ and $\mathbf{v}_1^{(r)}, \ldots, \mathbf{v}_K^{(r)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\text{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\text{ref}}^\intercal \mathbf{U}$,

$$\hat{\pi} = \underset{\pi \in \mathfrak{S}_R}{\arg\max} \sum_{h=1}^{R} |C_{r,\pi(r)}|, \qquad s_r = \operatorname{sign}\big(C_{r,\hat{\pi}(r)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_r, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(r)}$ are then obtained from the empirical distribution of the aligned draws.

We can deduce the following algorithm:

Algorithm 1 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$; dimension H; number of iterations T; burn-in T_0; hyperparameters a_0, b_0, λ

Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \ldots, T$ **do**
  $\mathbf{F}_{\mathbf{U}} \leftarrow \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of vMF($\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}}$); $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \ldots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of vMF($\mathbf{F}_{\mathbf{V}_k}$)
    for $h = 1, \ldots, H$ **do**
      $d_k^{(h)} \leftarrow \mathbf{u}^{(h)\intercal} \mathbf{X}_k \mathbf{v}_k^{(h)}$; draw $\sigma_k^{(h)} \sim \mathcal{N}^+\big(\tau_k d_k^{(h)}/(\tau_k+\lambda),\, 1/(\tau_k+\lambda)\big)$
    end for
    draw $\tau_k \sim \text{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K, \tau_1, \ldots, \tau_K\big)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 3.1)
return the aligned draws

4 Sparse Bayesian Framework for canonical factorisation

In OMCOA, each block is approximated as $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where the diagonal matrix $\Sigma_k = \text{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$ measures how much each latent dimension $\mathbf{u}^{(r)}$ contributes to block k. OMCOA always returns strictly positive weights, so that this pattern has to be assessed afterwards. We propose a Bayesian version of OMCOA in which a spike-and-slab prior on the weights $\sigma_k^{(r)}$ allows them to be exactly zero. The dimension R is then a *maximal* rank: inactive dimensions are switched off by the sampler, and the effective rank is inferred from the data.

The sparsity of the model is based on the diagonal terms $\sigma_k^{(r)}$ as follows

$$\sigma_k^{(r)} = \gamma_k^{(r)}\, s_k^{(r)}, \qquad \gamma_k^{(r)} \in \{0,1\}, \quad s_k^{(r)} > 0. \tag{5}$$

Priors. The difference between the two sets of prior distributions lies in the prior distributions of the diagonal terms:

- spike-and-slab on the weights: $\gamma_k^{(r)} \mid \theta_r \sim \text{Bernoulli}(\theta_r)$ and $s_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, i.e.

$$p(\sigma_k^{(r)} \mid \theta_r) = (1-\theta_r)\,\delta_0(\sigma_k^{(r)}) + \theta_r\,\mathcal{N}^+(\sigma_k^{(r)}; 0, \lambda^{-1});$$

- $\theta_r \sim \text{Beta}(a_\theta, b_\theta)$, one inclusion probability per dimension, shared by the K blocks.

Sharing θ_r across blocks borrows strength: a dimension that is active in several blocks is more likely to be active in another one, and a dimension that is inactive everywhere sees θ_r shrink towards zero.

As in the non-sparse case, the orthonormality of $\mathbf{U}$ and $\mathbf{V}_k$ gives

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_{r=1}^{R}\sigma_k^{(r)}d_k^{(r)} + \sum_{r=1}^{R}(\sigma_k^{(r)})^2, \qquad d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}. \tag{6}$$

Proposition 4 (Full conditionals)**.** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$.*

1. ***Common space:*** $\tilde{\mathbf{U}} \mid \cdot \sim \text{vMF}(\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})$ *with* $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$.
2. ***Block loadings:*** $\mathbf{V}_k \mid \cdot \sim \text{vMF}(\mathbf{F}_{\mathbf{V}_k})$ *with* $\mathbf{F}_{\mathbf{V}_k} = \tau_k\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. *When $\sigma_k^{(r)} = 0$, the r-th column of $\mathbf{F}_{\mathbf{V}_k}$ is zero and $\mathbf{v}_k^{(r)}$ is uniform on the orthogonal complement of the other columns.*
3. ***Weights (joint update of $\gamma_k^{(r)}$ and $\sigma_k^{(r)}$).*** *Let*

$$\omega_k = \tau_k + \lambda, \qquad m_k^{(r)} = \frac{\tau_k\,d_k^{(r)}}{\omega_k}.$$

Then, with $\sigma_k^{(r)}$ integrated out,

$$P(\gamma_k^{(r)} = 1 \mid \cdot) = \frac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}, \qquad B_k^{(r)} = 2\sqrt{\frac{\lambda}{\omega_k}}\,\exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big)\,\Phi\Big(m_k^{(r)}\sqrt{\omega_k}\Big),$$

where Φ is the standard normal c.d.f.; and

$$\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \cdot \;\sim\; \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1}), \qquad \sigma_k^{(r)} \mid \gamma_k^{(r)} = 0, \cdot \;=\; 0.$$

4. ***Inclusion probabilities:*** $\theta_r \mid \cdot \sim \text{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\; b_\theta + K - \sum_k \gamma_k^{(r)}\big)$.
5. ***Noise precisions:*** $\tau_k \mid \cdot \sim \text{Gamma}\big(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.

The matrix von Mises–Fisher distributions are sampled by one column-wise sweep per iteration, each column being drawn from a vector von Mises–Fisher distribution with Wood's algorithm, as in the non-sparse sampler. Updating $(\gamma_k^{(r)}, \sigma_k^{(r)})$ jointly, with $\sigma_k^{(r)}$ integrated out when drawing $\gamma_k^{(r)}$, is essential: drawing $\gamma_k^{(r)}$ given the current $\sigma_k^{(r)}$ would never allow the chain to leave $\sigma_k^{(r)} = 0$. In practice $\log B_k^{(r)}$ is computed on the log scale, with $\log\Phi$ evaluated directly, to avoid overflow.

Moreover, $B_k^{(r)}$ contains the factor $\sqrt{\lambda/\omega_k}$: a very diffuse slab ($\lambda \to 0$) always favours the spike (Lindley's paradox). The slab precision must therefore be chosen on the scale of the data, e.g. $\lambda = 1$ when the blocks are scaled so that $\|\mathbf{X}_k\|^2$ is of order one, which is the case when each block is divided by its Frobenius norm, as is usual in multiblock analysis.

From the posterior distributions of the parameter $\gamma_k^{(r)}$ a decision rule for the representation of the block $\mathbf{X}_k$ on the component $u^{(r)}$:

$$\text{the component } \mathbf{u}^{(r)} \text{ is active for the block } k \text{ if and only if } P(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \ldots, \mathbf{X}_K) > 1/2.$$

Moreover, this model makes it possible to estimate the rank of the blocks $(\mathbf{X}_1, ..., \mathbf{X}_K)$ i.e.

$$R_{\text{eff}} = \#\{r : \sum_k \gamma_k^{(r)} > 0\},$$

whose posterior distribution is estimated by its empirical distribution over the draws. The prior $\text{Beta}(a_\theta, b_\theta)$ controls the expected sparsity; $a_\theta = b_\theta = 1$ is a neutral choice, and $b_\theta > a_\theta$ favours sparser structures.

We can deduce Sparse Gibbs sampler algorithm for OMCOA (algorithm 2).

Algorithm 2 Sparse Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \ldots, \mathbf{X}_K$; maximal rank $R \le \min_k p_k$; iterations T; burn-in T_0; hyperparameters $a_0, b_0, \lambda, a_\theta, b_\theta$

Initialization: $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with R dimensions; $\gamma_k^{(r)} \leftarrow 1$; $\theta_r \leftarrow 1/2$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \ldots, T$ **do**
  update $\tilde{\mathbf{U}}$ by one sweep of $\text{vMF}\big(\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k\big)$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \ldots, K$ **do**
    update $\mathbf{V}_k$ by one sweep of $\text{vMF}\big(\tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k\big)$
    for $r = 1, \ldots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$, $\omega_k \leftarrow \tau_k + \lambda$, $m_k^{(r)} \leftarrow \tau_k d_k^{(r)} / \omega_k$
      draw $\gamma_k^{(r)} \sim \text{Bernoulli}\Big(\dfrac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}\Big)$
      if $\gamma_k^{(r)} = 1$ **then** draw $\sigma_k^{(r)} \sim \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1})$ **else** $\sigma_k^{(r)} \leftarrow 0$
    end for
    draw $\tau_k \sim \text{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  for $r = 1, \ldots, R$ **do**
    draw $\theta_r \sim \text{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\ b_\theta + K - \sum_k \gamma_k^{(r)}\big)$
  end for
  if $t > T_0$ **then** store $(\mathbf{U}, \Sigma_1, \ldots, \Sigma_K, \boldsymbol{\gamma}, \boldsymbol{\theta}, \boldsymbol{\tau})$
  end if
end for
align the stored draws (permutation and signs) on a reference $\mathbf{U}_{\text{ref}}$
return the aligned draws

5 Empirical performance via simulation

5.1 Simulation design

We consider $K = 3$ blocks with $p_1 = 10$, $p_2 = 5$, $p_3 = 15$ variables and $N = 100$ observations generated from

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad k = 1, 2, 3,$$

where $\mathbf{U} \in \mathcal{V}_{N,3}$ is column-centered, $\mathbf{V}_k \in \mathcal{V}_{p_k,3}$, Σ_k is the diagonal matrix of the block weights and $\mathbf{E}_k$ is a noise term. The true effective rank is $R_{\text{eff}} = 3$. The diagonal matrices are set to

$$\Sigma_1 = \text{diag}(0.7, 0.5, 0), \quad \Sigma_2 = \text{diag}(0.5, 0.7, 0), \quad \Sigma_3 = \text{diag}(0.3, 0, 0.5).$$

All the models are fitted with the maximal rank $R_{\max} = 5$, i.e. the largest value allowed by $R_{\max} \leq \min_k p_k$.

Grid for noise variance: We assume that the terms $e_{\bullet\bullet}$ of the noise $\mathbf{E}_k$ are distributed from $\mathcal{N}(0, s^2)$. For a spectral method applied in a single block and for any sufficiently large N, an active weight w is indetectable when the noise parameter s exceeds $w/(Np_k)^{1/4}$ (see Benaych-Georges and Nadakuditi, 2012, for details). Table 1 provides these thresholds as a function of the dimensional weights of the blocks. We set s within the interval from 0.02 to 0.18. Note that, for example, the first column of $\mathbf{U}$ remains probably well estimated even if $s > 0.05$ since it depends on the three blocks. On the other side, the third column of $\mathbf{U}$ depends only on block 3 so it will not be well estimated for $s > 0.08$.

dimension (r)	block (k)	$\sigma_k^{(r)}$	threshold
1	3	0.30	0.05
3	3	0.50	0.08
2	1	0.50	0.09
1	2	0.50	0.11
1	1	0.70	0.12
2	2	0.70	0.15

Table 1: Thresholds for the standard deviation of noise that make the weight of the dimension on the block undetectable.

Slab parameter λ: The slab scale $s_0 = \lambda^{-1/2}$ gives the expected order of magnitude of a nonzero weight $\sigma_k^{(r)}$ on the data scale: under the slab, the mean of $\sigma_k^{(r)}$ is $s_0\sqrt{2/\pi}$. When the blocks are standardized ($\|\mathbf{X}_k\| = 1$), the weights are bounded, $\sigma_k^{(r)} \leq 1$, and $(\sigma_k^{(r)})^2$ is the proportion of the variance of block k explained by dimension r. With $\lambda = 1$, the slab puts 68% of its mass on $[0, 1]$ and allows a single dimension to explain most of the variance of a block. With $\lambda = 0.1$ ($s_0 \approx 3.2$), only 25% of its mass lies on attainable values, which penalizes the slab against the spike and makes the selection more conservative. With $\lambda = 10$ ($s_0 \approx 0.32$), the slab is concentrated on small weights, which favours the inclusion of weak dimensions.

The gibbs sampler was built with $T = 6000$ iterations, a burn-in of $T_0 = 1000$. For the sparse version, the parameter λ varied in $\{0.1, 1, 10\}$ and $a_\theta = b_\theta = 1$. All the fitted configuration $\hat{U}$ were aligned on the true $\mathbf{U}$ (permutation and signs of the dimensions). $B = 1000$ data sets

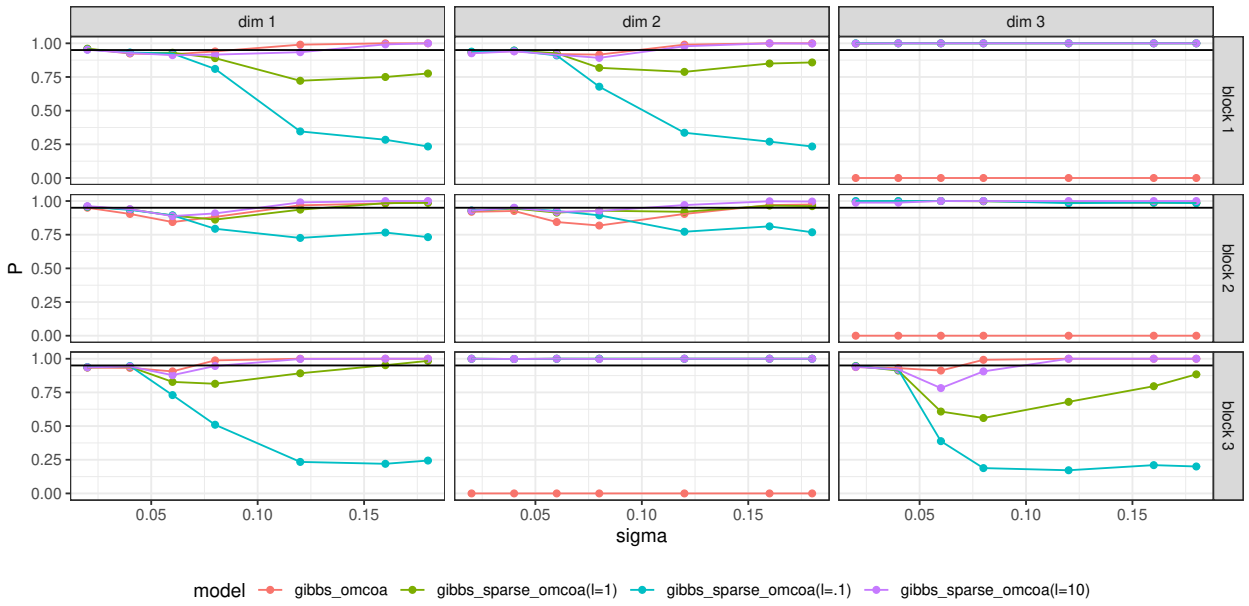


Figure 1: Empirical coverage probability for gibbs-omcoa (in red), gibbs-sparse-omcoa (green for $\lambda = 1$, blue for $\lambda = .1$ and purple for $\lambda = 10$) as a function of the noise parameter σ.

were simulated in each case. For the effective rank, the estimate obtained using the Gibbs sparse sampler are compared with the estimate of the Bayasian SVD of Hoff on the concatenated block $\mathbf{X} = [\mathbf{X}_1|...|\mathbf{X}_K]$ (see Hoff, 2007, for details). The effective rank estimate is also compared to the rank obtained using the Gavish-Donoho threshold when the noise variance is unknown (see Gavish and Donoho, 2014, for details).

5.2 Results

Figure 1 provides the estimated block weights and their 95%-CI of the active dimensions (dimension 1 to 3) as function of the proportion σ associated to the residual error . When the weigths were non null, they were accurately estimated using both bayesian frameworks and the omcoa algorithm for all values of σ.The bias estimation of these weights with the sequential approach (mcoa) can be increased with σ (for example the variance of the block 2 explained by dimension 1). When the weights were null, as expected, the bias of the sparse version remained near from 0 for all σ, whereas the bias of the others increased with σ. We can note that estimates obtained using the Gibbs sampler degrade less than those obtained using mcoa and omcoa methods.

Figure 2 provides the estimated block weights and their 95%-CI of the inactive dimensions (dimension 4 and 5) as function of the proportion σ associated to the residual error . The bias of estimate of the sequential approach (mcoa, omcoa) and the gibbs sampler increased with σ. For the sparse version, the bias remained to 0 in all cases.

Analysis of sensitivity to λ: We compare the performances in terms of sensitivity (the proportion of active dimensions that are selected) and of specificity (proportion of inactive dimensions that are not selected) of the decision rule defined in (4). The selection rule had a very high empirical

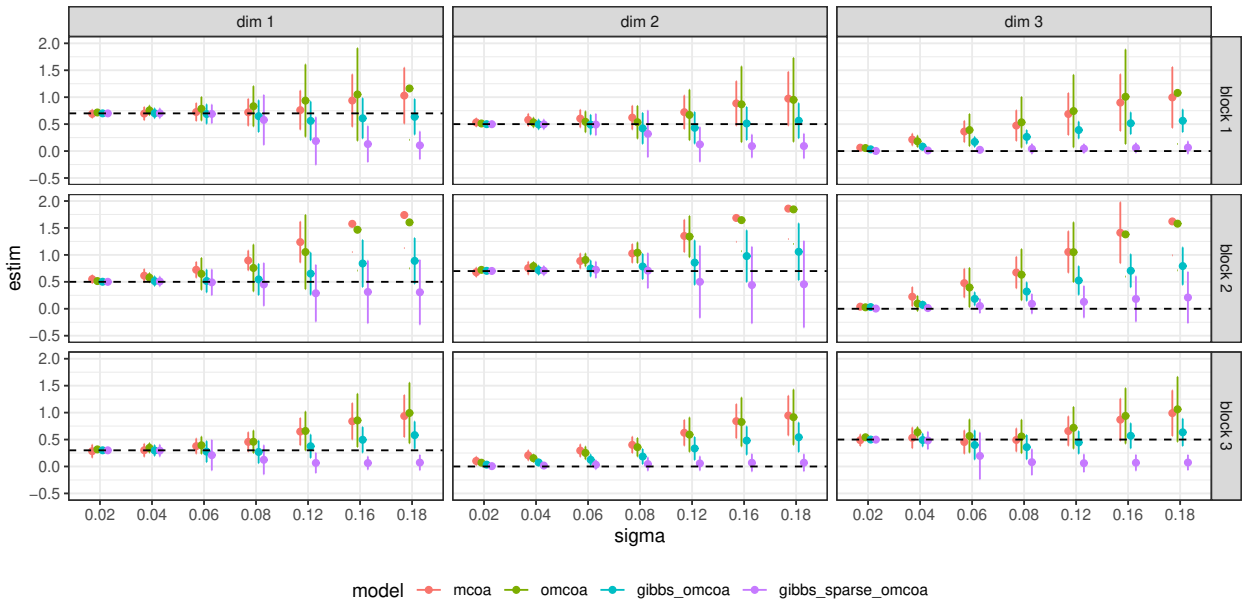


Figure 2: Estimated block weights and their 95% empirical confidence intervals for the three active dimensions (dimensions 1 through 3) obtained using the sequential approach (MCOA in red), the simultaneous approach (OMCOA in green), the Gibbs sampler for OMCOA (Gibbs OMCOA in light blue) and the sparse version (sparse Gibbs OMCOA in purple), as a function of the noise standard deviation s. For Bayesian models, the estimate is equal to the posterior mean.

sensitivity : 100% for all σ. Its empirical specificity decreased as a function of σ but remainded very high: exceeding 99.4% for $\lambda = 0.1$, above 98.3% for $\lambda = 1$ and above 92.7% for $\lambda = 10$. Figure 3 shows the empirical specificity at block and dimension levels. The specificity for block 2 with dimensions 4 and 5 can be linked to the estimated values obtained for the block weights (Figure 2). Figure 4 displays the proportion of dimensions chosen. The posterior mode of R_{eff} equals the true value of 3 in 95.4% of cases on average for σ with $\lambda = 0.1$ and in 85.7% of cases on average for σ with $\lambda = 1$. For $\lambda = 10$, this proportion is only 0.404, the choice $R_{\text{eff}} = 4$ is the majoritory choice.

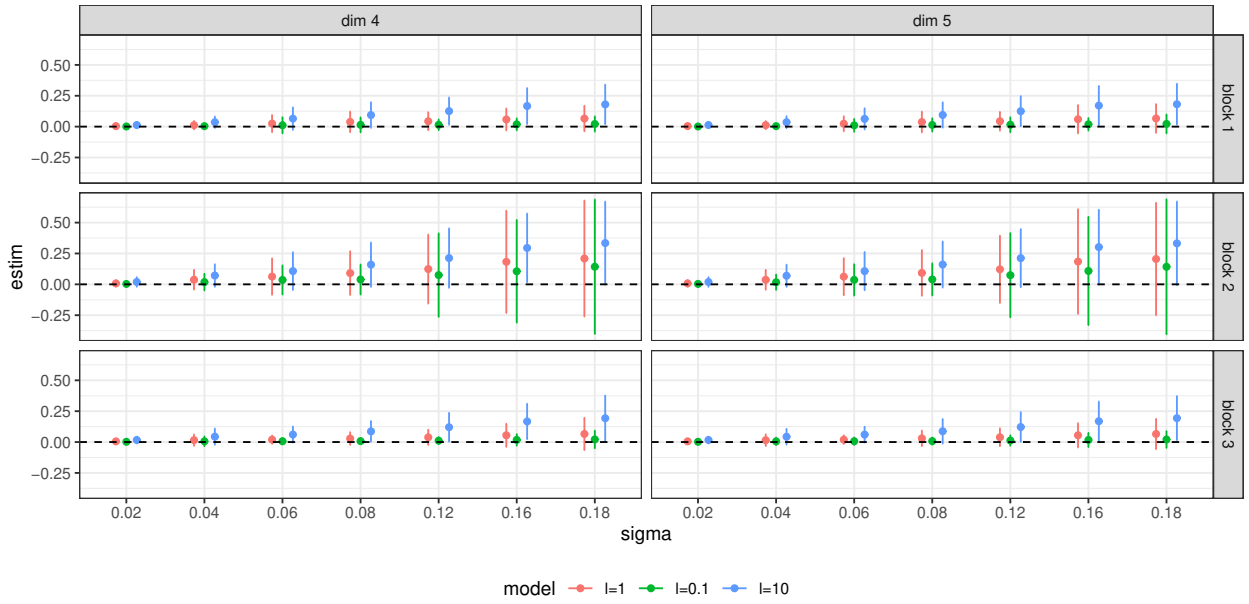


Figure 3: Estimated block weights and their empirical 95%-CI of the two non actives dimensions (dimension 4 and 5) using the the sparse version (gibbs sparse omcoa) as a function of the standard deviation s of the noise and the slab parameter λ.

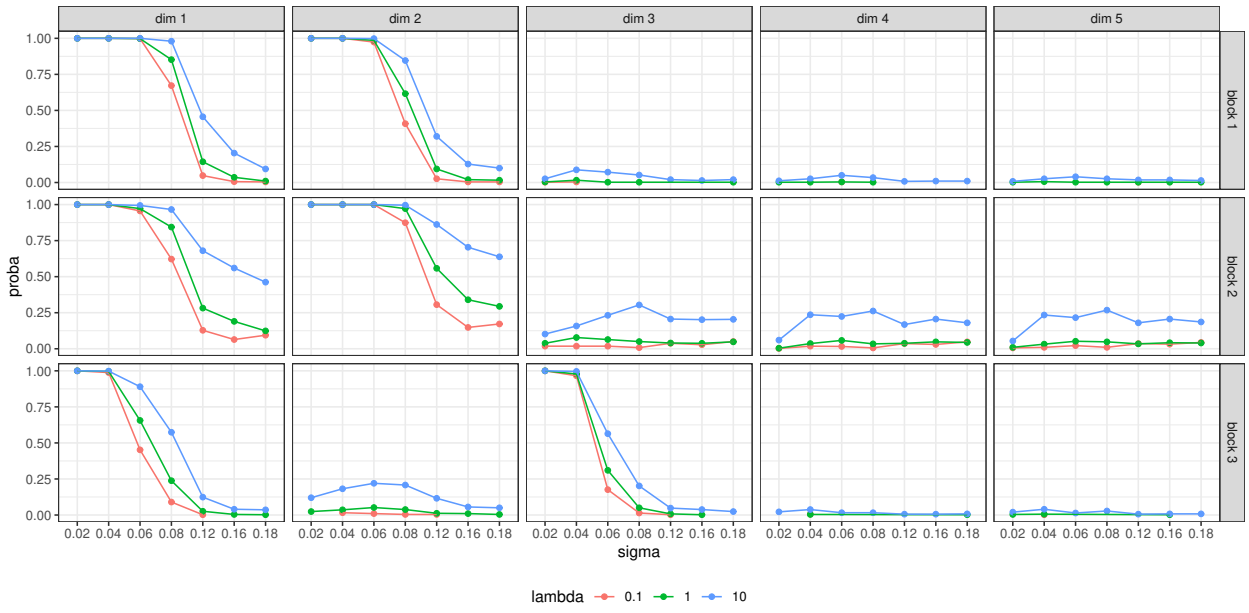


Figure 4: Empirical probability of detection an active dimension in a block as function of the standard deviation s of the residual error of the slab precision λ.

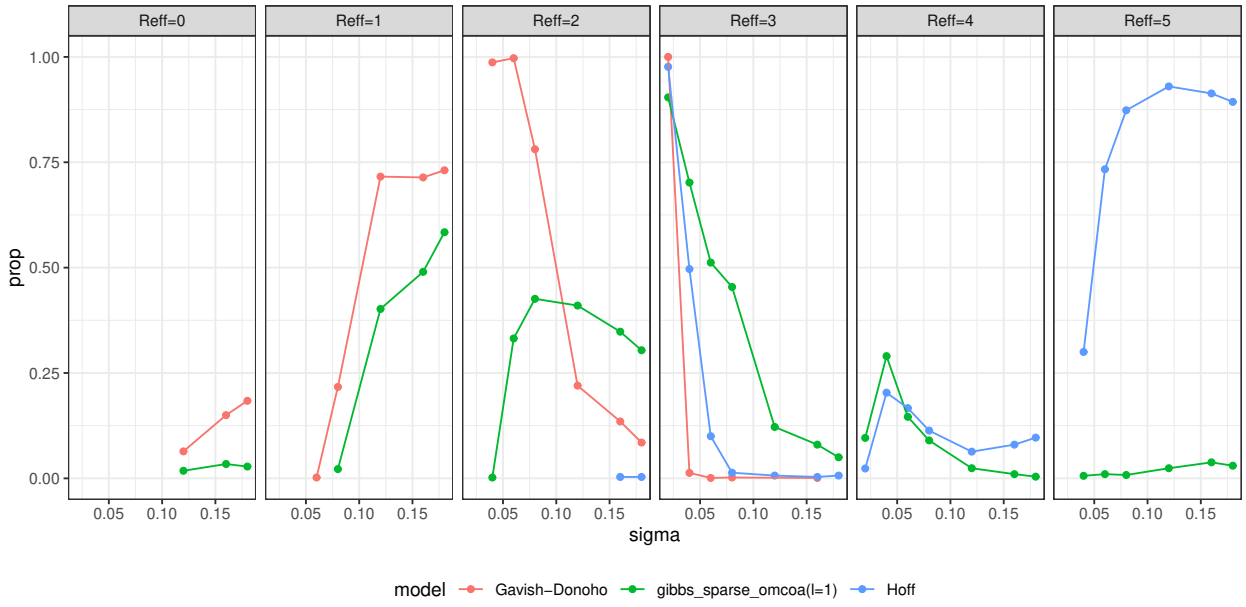


Figure 5: Proportion of the number of dimension chosen as function of the residual parameter σ and of the parameter λ.

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Appendix: proof of the proposition

Proposition 1. The criterion to be minimized is:

$$\mathcal{L}(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K) = \sum_{k=1}^{K} \left\| \mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\top \right\|^2,$$

subject to $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all $k = 1, \ldots, K$.

The algorithm is of the *Alternating Least Squares* (ALS) type: at each iteration, the criterion is minimized successively with respect to each parameter, while the others are held fixed. It therefore suffices to show that each individual update decreases or maintains $\mathcal{L}$.

Update of U With $\mathbf{V}_k$ and Σ_k fixed, let $\mathbf{H}_k = \Sigma_k \mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1 \mid \cdots \mid \mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. Then:

$$\mathcal{L} = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\mathbf{H}_k\|^2 = \|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2.$$

Expanding: $\|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2 = \|\mathbf{X}\|^2 - 2\ \mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) + \|\mathbf{U}\mathbf{H}^\intercal\|^2$.

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$: $\|\mathbf{U}\mathbf{H}^\intercal\|^2 = \mathrm{trace}(\mathbf{H}\mathbf{U}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}\mathbf{H}^\intercal) = \|\mathbf{H}\|^2$ (constant w.r.t. $\mathbf{U}$).

Minimizing $\mathcal{L}$ with respect to $\mathbf{U}$ is therefore equivalent to maximizing

$$\mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}^\intercal\mathbf{X}^\intercal\mathbf{U}) = \mathrm{trace}(\mathbf{U}^\intercal\mathbf{X}\mathbf{H}),$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{A}\mathbf{B}) = \mathrm{trace}(\mathbf{B}\mathbf{A})$ for $\mathbf{A}, \mathbf{B}$ of dimensions $N \times P$ and $P \times N$. The second equality comes from the equality of the trace of a square matrix and its transpose. By Cliff (1966), the maximum is attained at $\mathbf{U}^* = \mathbf{A}\mathbf{B}^\intercal$, where $\mathbf{X}\mathbf{H}^\intercal = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the thin SVD of $\mathbf{X}\mathbf{H}^\intercal \in \mathcal{M}_{N,R}(\mathbb{R})$. This update is the global minimizer of $\mathcal{L}$ with respect to $\mathbf{U}$, hence:

$$\mathcal{L}(\mathbf{U}^*, \ldots) \leq \mathcal{L}(\mathbf{U}_{\text{old}}, \ldots).$$

Update of $\mathbf{V}_k$ With $\mathbf{U}$ and Σ_k fixed, expanding the contribution of block k:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) + \|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2.$$

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$:

$$\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \mathrm{trace}(\mathbf{V}_k\Sigma_k^\intercal\mathbf{U}^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\Sigma_k^2) \quad \text{(constant w.r.t. } \mathbf{V}_k\text{)}.$$

Minimizing therefore reduces to maximizing $\mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k)$ subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$, which requires $R \leq p_k$. By Cliff (1966), the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the thin SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$. Note that Σ_k cannot be dropped: in general the polar factor of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ differs from that of $\mathbf{X}_k^\intercal\mathbf{U}$. If some $\sigma_k^{(r)}$ are zero, $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ is rank deficient and the maximizer is not unique, but any maximizer gives the same value of the criterion. Hence:

$$\mathcal{L}(\ldots, \mathbf{V}_k^*, \ldots) \leq \mathcal{L}(\ldots, \mathbf{V}_{k,\text{old}}, \ldots).$$

Update of Σ_k With $\mathbf{U}$ and $\mathbf{V}_k$ fixed (both having orthonormal columns), and $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, the contribution of block k is:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k) + \mathrm{trace}(\Sigma_k^2).$$

Since Σ_k is diagonal, it remains to minimize:

$$\sum_{r=1}^{R} \Big[(\sigma_k^{(r)})^2 - 2\,\sigma_k^{(r)}\,(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr} \Big].$$

Each term is a convex parabola in $\sigma_k^{(r)}$, minimized at: $(\sigma_k^{(r)})^* = (\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)_{rr}$, or, in matrix form:$\Sigma_k^* = \mathrm{diag}(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)$. This update is therefore the exact minimizer with respect to Σ_k. If $(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)_{rr} < 0$, replacing $\mathbf{v}_k^{(r)}$ by $-\mathbf{v}_k^{(r)}$ and $\sigma_k^{(r)}$ by $|(\mathbf{U}^\intercal \mathbf{X}_k \mathbf{V}_k)_{rr}|$ keeps $\mathbf{V}_k$ orthonormal and leaves the criterion unchanged, so that the constraint $\sigma_k^{(r)} \geq 0$ costs nothing. Hence:

$$\mathcal{L}(\ldots, \Sigma_k^*, \ldots) \leq \mathcal{L}(\ldots, \Sigma_{k,\mathrm{old}}, \ldots).$$

Global Convergence At each complete iteration (update of $\mathbf{U}$, then all $\mathbf{V}_k$, then all Σ_k), the criterion $\mathcal{L}$ is monotonically decreasing and bounded below by zero. A monotonous sequence bounded below converges, so the sequence of values $(\mathcal{L}^{(t)})_{t\geq 0}$ converges:

$$\mathcal{L}^{(t+1)} \leq \mathcal{L}^{(t)}, \quad \mathcal{L}^{(t)} \geq 0 \implies \mathcal{L}^{(t)} \xrightarrow[t\to\infty]{} \ell^* \geq 0.$$

□

Proof. (a) The true parameters give the value 0 to criterion (2), so every minimiser $(\mathbf{U}', \mathbf{V}'_k, \Sigma'_k)$ satisfies $\mathbf{X}_k = \mathbf{U}'\Sigma'_k\mathbf{V}'^\intercal_k$ for all k; this does not depend on the weights given to the blocks. Since every dimension is present in some block, the column space of $\mathbf{X} = [\mathbf{X}_1|\cdots|\mathbf{X}_K]$ is the span of $\mathbf{U}$, of dimension R. It is contained in the span of $\mathbf{U}'$, which also has dimension R; hence $\mathbf{U}' = \mathbf{U}\mathbf{O}$ with $\mathbf{O}$ orthogonal. From $\mathbf{X}_k\mathbf{X}_k^\intercal = \mathbf{U}\Sigma_k^2\mathbf{U}^\intercal = \mathbf{U}'\Sigma_k'^2\mathbf{U}'^\intercal$ we get

$$\mathbf{O}^\intercal \Sigma_k^2 \mathbf{O} = \Sigma_k'^2, \qquad k = 1, \ldots, K.$$

For $\mathbf{c} \in \mathbb{R}^K$, the diagonal matrix $\mathbf{A}(\mathbf{c}) = \sum_k c_k \Sigma_k^2$ has entries $a_r = \sum_k c_k (\sigma_k^{(r)})^2$, and $\mathbf{O}^\intercal \mathbf{A}(\mathbf{c}) \mathbf{O}$ is diagonal. Since the profiles are distinct and nonnegative, the squared profiles are distinct, and $\mathbf{c}$ can be chosen outside the finite union of hyperplanes $\{a_r = a_{r'}\}$, so that the entries of $\mathbf{A}(\mathbf{c})$ are pairwise distinct. An orthogonal matrix that diagonalises a diagonal matrix with distinct entries is a signed permutation matrix: $\mathbf{O} = \mathbf{P}\mathbf{D}$, with $\mathbf{P}$ a permutation matrix and $\mathbf{D} = \mathrm{diag}(\pm 1)$. Hence $\mathbf{U}' = \mathbf{U}\mathbf{P}\mathbf{D}$ and $\Sigma_k'^2 = \mathbf{P}^\intercal \Sigma_k^2 \mathbf{P}$, that is $\Sigma'_k = \mathbf{P}^\intercal \Sigma_k \mathbf{P}$ since the weights are nonnegative. Finally, $\mathbf{U}'^\intercal \mathbf{X}_k = \Sigma'_k \mathbf{V}'^\intercal_k$ gives, for every (k, r) with $\sigma_k'^{(r)} > 0$, $\mathbf{v}_k'^{(r)} = \mathbf{X}_k^\intercal \mathbf{u}'^{(r)} / \sigma_k'^{(r)}$, which is the matching column of $\mathbf{V}_k$ with the sign of $\mathbf{u}'^{(r)}$; the columns with $\sigma_k'^{(r)} = 0$ are only constrained by $\mathbf{V}'_k \in \mathcal{V}_{p_k,R}$.

(b) Write $\mathcal{R}_t$ for the set of dimensions not yet extracted at step t of MCOA ($\mathcal{R}_1 = \{1, \ldots, R\}$), and assume that the deflated blocks are $\mathbf{X}_k^{(t)} = \sqrt{w_k} \sum_{r\in\mathcal{R}_t} \sigma_k^{(r)} \mathbf{u}^{(r)} \mathbf{v}_k^{(r)\intercal}$, which holds for $t = 1$. Then

$$\mathbf{X}^{(t)}\mathbf{X}^{(t)\intercal} = \sum_k \mathbf{X}_k^{(t)}\mathbf{X}_k^{(t)\intercal} = \sum_{r\in\mathcal{R}_t} \lambda_r(\mathbf{w})\, \mathbf{u}^{(r)}\mathbf{u}^{(r)\intercal},$$

because the columns of each $\mathbf{V}_k$ are orthonormal. The values $\lambda_r(\mathbf{w})$ being distinct, the leading left singular vector of $\mathbf{X}^{(t)}$ is $\pm\mathbf{u}^{(r_t)}$, with $r_t = \arg\max_{r\in\mathcal{R}_t} \lambda_r(\mathbf{w})$. Then $\mathbf{X}_k^{(t)\intercal}\mathbf{u}^{(r_t)} = \sqrt{w_k}\, \sigma_k^{(r_t)} \mathbf{v}_k^{(r_t)}$: the MCOA weight is $\sqrt{w_k}\, \sigma_k^{(r_t)}$ and, when it is positive, the normalised loading is $\pm\mathbf{v}_k^{(r_t)}$ (with the sign of $\mathbf{u}$), so that the deflation $\mathbf{X}_k^{(t)}(\mathbf{I} - \mathbf{v}_k^{(r_t)}\mathbf{v}_k^{(r_t)\intercal})$ removes exactly the term r_t. When the weight is zero, the term r_t is absent from block k and the block is left unchanged. The induction hypothesis holds at step $t+1$ with $\mathcal{R}_{t+1} = \mathcal{R}_t \setminus \{r_t\}$. After R steps, MCOA has recovered $\mathbf{U}$ and the $\mathbf{V}_k$ (columns with a positive weight) up to order and signs, and gives a zero value to criterion (2) for the weighted blocks: by (a), it coincides with the OMCOA solution.

(c) Let t be the first step where $\max_{r\in\mathcal{R}_t} \lambda_r(\mathbf{w})$ is attained by several dimensions $r \in \mathcal{T} \subset \mathcal{R}_t$. Up to step $t-1$, the argument of (b) applies, and the leading eigenvalue of $\mathbf{X}^{(t)}\mathbf{X}^{(t)\intercal}$ has the

eigenspace $\mathrm{span}\{\mathbf{u}^{(r)}, r \in \mathcal{T}\}$, of dimension $|\mathcal{T}| \geq 2$. Every unit vector of this space maximises the MCOA criterion at step t; a vector other than $\pm\mathbf{u}^{(r)}$, $r \in \mathcal{T}$, gives a common component that is not a column of $\mathbf{U}$ up to sign, hence, by (a), a solution different from the OMCOA one. □

Proposition 2. (1)–(2): by (4), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k \,\mathrm{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U})\} = \exp\{\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{U})\}$, and $\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal \mathbf{Q}\tilde{\mathbf{U}}) = \mathrm{trace}\big((\mathbf{Q}^\intercal \mathbf{F}_\mathbf{U})^\intercal \tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\mathrm{trace}(\Sigma_k \mathbf{V}_k^\intercal \mathbf{X}_k^\intercal \mathbf{U}) = \mathrm{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal \mathbf{V}_k)/\tau_k$. (3): the terms of the log-likelihood involving $\sigma_k^{(r)}$ are, by (4), $-\frac{\tau_k}{2}\big[(\sigma_k^{(r)})^2 - 2\sigma_k^{(r)} d_k^{(r)}\big]$; adding the log-prior $-\frac{\lambda}{2}(\sigma_k^{(r)})^2$ and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(r)} > 0$. (4): the Gamma prior is conjugate for the precision of a Gaussian likelihood. □

Proposition 3. Items 1, 2, 4 and 5 follow as in the non-sparse sampler (linearity of (4) in $\mathbf{U}$ and $\mathbf{V}_k$, and conjugacy). For item 3, the terms of the log-likelihood involving $\sigma_k^{(r)}$ are, by (6), $\tau_k d_k^{(r)} \sigma_k^{(r)} - \frac{\tau_k}{2}(\sigma_k^{(r)})^2$. Under $\gamma_k^{(r)} = 0$, $\sigma_k^{(r)} = 0$ and this quantity is 0. Under $\gamma_k^{(r)} = 1$, integrating against the half-normal slab of density $2\sqrt{\lambda/(2\pi)}\exp(-\lambda s^2/2)$ on $(0, \infty)$ gives

$$2\sqrt{\frac{\lambda}{2\pi}} \int_0^\infty \exp\Big(-\frac{\omega_k}{2} s^2 + \tau_k d_k^{(r)} s\Big) ds = 2\sqrt{\frac{\lambda}{\omega_k}} \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \Phi\big(m_k^{(r)} \sqrt{\omega_k}\big) = B_k^{(r)},$$

which is the Bayes factor of $\gamma_k^{(r)} = 1$ against $\gamma_k^{(r)} = 0$. The conditional of $\sigma_k^{(r)}$ given $\gamma_k^{(r)} = 1$ is obtained by completing the square. □